C. V. Raman Global University
Odisha

R25

Registration No. :

Total number of printed pages – 4

B.Tech
CS242

THIRD SEMESTER REGULAR EXAMINATION – 2025

DATABASE ENGINEERING

CSE & ALLIED BRANCHES

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Explain recursive relationship with an example.	1	2	2
(b)	Differentiate between strong entity sets and weak entity sets in a database. Explain their characteristics, key dependencies, and provide suitable examples to illustrate the distinction.	1	2	2
(c)	Highlight the important tasks handled by a Database Administrator (DBA) in a Database Management System.	1	2	1
(d)	Explain the key terminologies of the Relational Model — Relation, Tuple, Domain, and Cardinality — and describe how each contributes to the structure of a relational database.	2	2	1
(e)	List any two Relational Model Constraints and describe their significance by giving suitable example of the same.	2	2	2
(f)	State the difference between simple vs multivalued, stored vs derived attribute with suitable examples.	2	2	2
(g)	Outline the Selection and Projection operations used in relational algebra with syntax.	3	2	2
(h)	Given Schema: Student (sid, sname, branch, address). Write the query using SQL to retrieve the names of students whose name is 7 characters long and the name starts with 'M'.	3	2	2
(i)	Compare and contrast TRUNCATE and DROP command of SQL with example.	3	2	2
(j)	Consider the relation R(STUVW) with the following set of functional dependencies: F = { ST → UVW, V → TW } Determine minimum two possible candidate keys for the relation.	4	2	3
(k)	Explain the difference between Partial Dependency and Transitive Dependency in the context of relational database design.	4	2	2
(l)	Explain Armstrong's Axioms and describe their role in inferring functional dependencies within a relational schema.	4	2	2
(m)	Compare conflict serializability and view serializability with suitable examples.	5	2	2

(n)	Compare conflict operations (read–write, write–read, write–write conflicts) with examples.	5	2	4
(o)	Illustrate and describe the various types of database failures that may affect transaction processing.	5	2	2

	Part-B	CO	Marks	Bl
Q2.	**Answer any two:**			
(a)	Explain the disadvantages of the traditional file processing system and the need for a Database Management System (DBMS).	1	7	1
(b)	Explain the major components of the Entity-Relationship (ER) model in detail. Describe the notations of entities, attributes, relationships, and keys. Explain the different types of relationships in an ER model with examples.	1	7	2
(c)	Illustrate and describe the three levels of database architecture using a suitable diagram. Explain how data abstraction enhances database efficiency and user interaction.	1	7	2
Q3.	**Answer any two:**			
(a)	Briefly explain the concepts of Generalization and Specialization in the Enhanced Entity–Relationship (EER) Model. Describe how these concepts help in representing hierarchical relationships among entities, and illustrate your explanation with suitable examples or diagrams.	2	7	3
(b)	i) Design an ER diagram for a *university database* that stores information about students, courses, and instructors. a. Each student can enrol in multiple courses. b. Each course can be taught by one instructor. c. Include appropriate attributes, relationships, and key constraints. ii) Map the ER diagram into relational tables following standard ER-to-Relational Mapping rules. Justify the decisions behind the design with specifying all keys and constraints clearly.	2	7	4
(c)	Discuss the different types of keys used in a Relational Database Management System. Explain the purpose of each key type — Super Key, Candidate Key, Primary Key, Alternate Key, and Foreign Key — with suitable examples to illustrate their role in maintaining data integrity.	2	7	
Q4.	**Answer any two:**			
(a)	Analyse the concept of Inner Join in Relational Algebra. Discuss how it operates on two relations to produce combined tuples based on a matching condition. Differentiate between natural join, theta join, and equi-join as forms of inner join. Apply your understanding by illustrating each type with relevant examples using suitable relational schemas	3	7	

(b)	Based on the following table answer all the questions by writing SQL queries. **University_Faculty_Details** *ID* | *name* | *salary* | *dept_name* | *building* | *budget* 22222 | Einstein | 95000 | Physics | Watson | 70000 12121 | Wu | 90000 | Finance | Painter | 120000 32343 | El Said | 60000 | History | Painter | 50000 45565 | Katz | 75000 | Comp. Sci. | Taylor | 100000 98345 | Kim | 80000 | Elec. Eng. | Taylor | 85000 76766 | Crick | 72000 | Biology | Watson | 90000 10101 | Srinivasan | 65000 | Comp. Sci. | Taylor | 100000 58583 | Califieri | 62000 | History | Painter | 50000 83821 | Brandt | 92000 | Comp. Sci. | Taylor | 100000 15151 | Mozart | 40000 | Music | Packard | 80000 33456 | Gold | 87000 | Physics | Watson | 70000 76543 | Singh | 80000 | Finance | Painter | 120000 1) Find the name, dept_name, salary of those faculty members whose salary is between ₹70,000 and ₹90,000. 2) List name, dept_name, salary of those faculties whose name starts with the letter 'S' 3) List all faculty names in ascending order of salary. 4) Update the salary of all faculties by 10% increment. 5) Display the ProductName and Price, sorted by ProductName in ascending order. 6) Add a new column "phone_number" to the above table. 7) Insert three new records into the above table	3	7
(c)	Write the following queries by using Relational Algebra product (*product_id*, product_name, manufacturer_name, product_rate, sell_price, product_description,client_id) client (*client_id*, client_name, address, city,pin , state ,bal_due) salesman (*salesman_id*, salesman_name,address, city, pin, state,salary,product_id) i) Find the names of salesmen and clients who are from same city. ii) Find all salesmen name, address pin and city who sell products with a selling price greater than ₹10,000. iii) Find the salesmen names who sell products manufactured by "Samsung". iv) Find the client details who lives in Odisha and balance due more than 20000. v) Find the client_id and names who sell Washing machines.	3	7

Q5.	Answer the followings:			
(a)	Describe the necessary conditions under which two sets of functional dependencies are considered equivalent. For the relation R(WXYZ), analyse the equivalence between the following two sets of functional dependencies: $FD_1 = \{ W \rightarrow X, X \rightarrow Y, WX \rightarrow Z \}$ $FD_2 = \{ W \rightarrow X, X \rightarrow Y, W \rightarrow Y, W \rightarrow Z \}$ Determine whether both sets are equivalent.	4	7	3
(b)	a) Explain normalization in DBMS. Discuss the different normal forms up to Third Normal Form (3NF) b) Explain the concept of lossless join decomposition in relational database design. Given a relational schema R = {SSN, ENAME, PNUMBER, PNAME, PLOCATION, HOURS,CITY,STATUS } with the FDs SSN →{ ENAME,CITY} PNUMBER → { PNAME, PLOCATION}, { SSN, PNUMBER} → HOURS}, {CITY->STATUS}. Verify if the given relation is in BCNF. If not, perform the necessary decomposition to achieve BCNF.	4	7	4
Q6.	**Answer the followings:**			
(a)	Elaborate on the concept of a transaction in a Database Management System (DBMS). Describe the ACID properties and various states of a transaction, and illustrate how they contribute to maintaining consistency and reliability in database operations with appropriate example.	5	7	2
(b)	Check which of the following schedule is conflict serializable: S1: r1(x);r3(x);w3(x);w1(x),r2(x) S2: r3(x);r2(x);r1(x);w3(x);w1(x) S3: r1(x); w2(x); r3(x); w1(x); r2(x); w3(x)	5	7	3

Best of Luck

C. V. Raman Global University
Odisha

R26

Registration No. :

Total number of printed pages – 3

B.Tech.
Subject Code: CY201

4th SEMESTER REGULAR EXAMINATION – 2026

SUBJECT NAME: Green Chemistry

BRANCH: CE, CH, AIML, ICB, CSIT, ME, MR

Time – 3 Hours

Full Marks – 100

Answer Questions from Part – A and Part – B

Q1.	Part-A (Answer all questions)	CO	Marks	BL
(a)	What is osmosis?	CO2	2	2
(b)	Define oxidation and reduction.	CO3	1 + 1	1,2
(c)	What is pitting corrosion?	CO5	2	2
(d)	If 20 g of KCl is present in 100 mL of water. Calculate its concentration in terms of ppm.	CO2	2	2
(e)	Write two examples of renewable source of energy.	CO1	1 + 1	2
(f)	'While iron, under exposure to oxygen and moisture undergoes corrosion, gold remains unaffected.' Explain.	CO5	2	1,2
(g)	Write the name of industrial process to synthesize ammonia from nitrogen and hydrogen.	CO1	2	2
(h)	What is desalination?	CO2	2	2,3
(i)	What is electrolyte used in 'proton exchange membrane fuel cell'?	CO4	2	2
(j)	Write the electrolyte of 'Direct Methanol Fuel Cell'?	CO4	2	1
(k)	How can you differentiate between alkaline battery and acid battery, giving suitable examples?	CO3	2	1,2
(l)	Write the name of the indicator used in EDTA titration to estimate hardness of water.	CO2	2	2

(m)	Write two examples of bio-fuel.	CO1	1+1	2
(n)	Which of the following salt is used to re-generate an exhausted zeolite bed? i) $CaSO_4$; ii) NaCl; iii) K_2SO_4; iii) $MgCl_2$.	CO2	2	2,3
(o)	Write two examples of Green Solvents.	CO1	2	2,3

	Part-B	CO	Marks	BL
Q2.	**Answer any two:**			
(a)	What is biocatalysis? Mention the 12 principles of green chemistry.	CO1	1+6	2
(b)	Calculate % of atom economy and E factor for the following two reactions Reaction 1 $Fe_2O_3 + 3\,CO \longrightarrow 2\,Fe + 2CO_2$ 159 28 56 44 Reaction 2 $C_2H_4 + H_2O \longrightarrow C_2H_6O$	CO1	3.5+3.5	3
(c)	With the help of a diagram, explain the principle of reverse osmosis.	CO2	7	4
Q3.	**Answer any two:**			
(a)	Illustrate softening of hardwater through Ion exchange method.	CO2	5 + 2	3,4
(b)	How much of $CaSO_4$ needed to be present in 1 litre of distilled water to have a hardness of 100 ppm. Molecular weight of $CaSO_4$ is 136.	CO2	7	3,
(c)	'While EBT in hardwater provides a wine-red colour, it changes to steel blue upon addition of EDTA'. Explain the colour change and the chemical process involved here.	CO2	7	4
Q4.	**Answer any two:**			
(a)	Explain the chemical reactions and origin of flow of electrons in a Daniel Cell having Zn as anode and Cu as cathode.	CO3	7	3,4
(b)	Write the cathode, anode and electrolyte of Li-ion battery. Describe the discharging and charging process from the prospective of intercalation of Li ions.	CO3	5 + 2	3,4

(c)	What is memory effect? Write the cathode and anode reactions of Ni-Cd Battery.	CO3	2+5	2,4
Q5.	**Answer the followings:**			
(a)	What is the electrolyte of an alkaline fuel cell? Write the anode, cathode and overall reaction of an alkaline fuel cell.	CO4	2+5	4
(b)	Determine the Pilling-Bedworth ratio in case of a MgO layer formation over Mg. M.W (MgO) = 41.31 g/mol; d (MgO) = 3.58 g/cm^3; A. W (Mg) = 24.3 and d (Mg) = 1.74 g/cm^3. Comment on the protective vs non-protective nature of MgO layer.	CO5	5+2	2,4
Q6.	**Answer the followings:**			
(a)	Explain the corrosion control method through Cathodic Protection.	CO5	3.5+3.5	2,4
(b)	What should be used as anode, cathode and electrolyte in case of copper coating on an iron rod? Explain the chemical process involved in coating process.	CO5	3+4	2,3

Best of Luck

07/04/26 - 2nd Sitting

C. V. Raman Global University
Odisha

B26

Registration No. : | | | | | | | | | | |

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BTECH
Subject Code: CSE23306

FOURTH SEMESTER BACKLOG EXAMINATION – 2026

SUBJECT NAME: Database Engineering

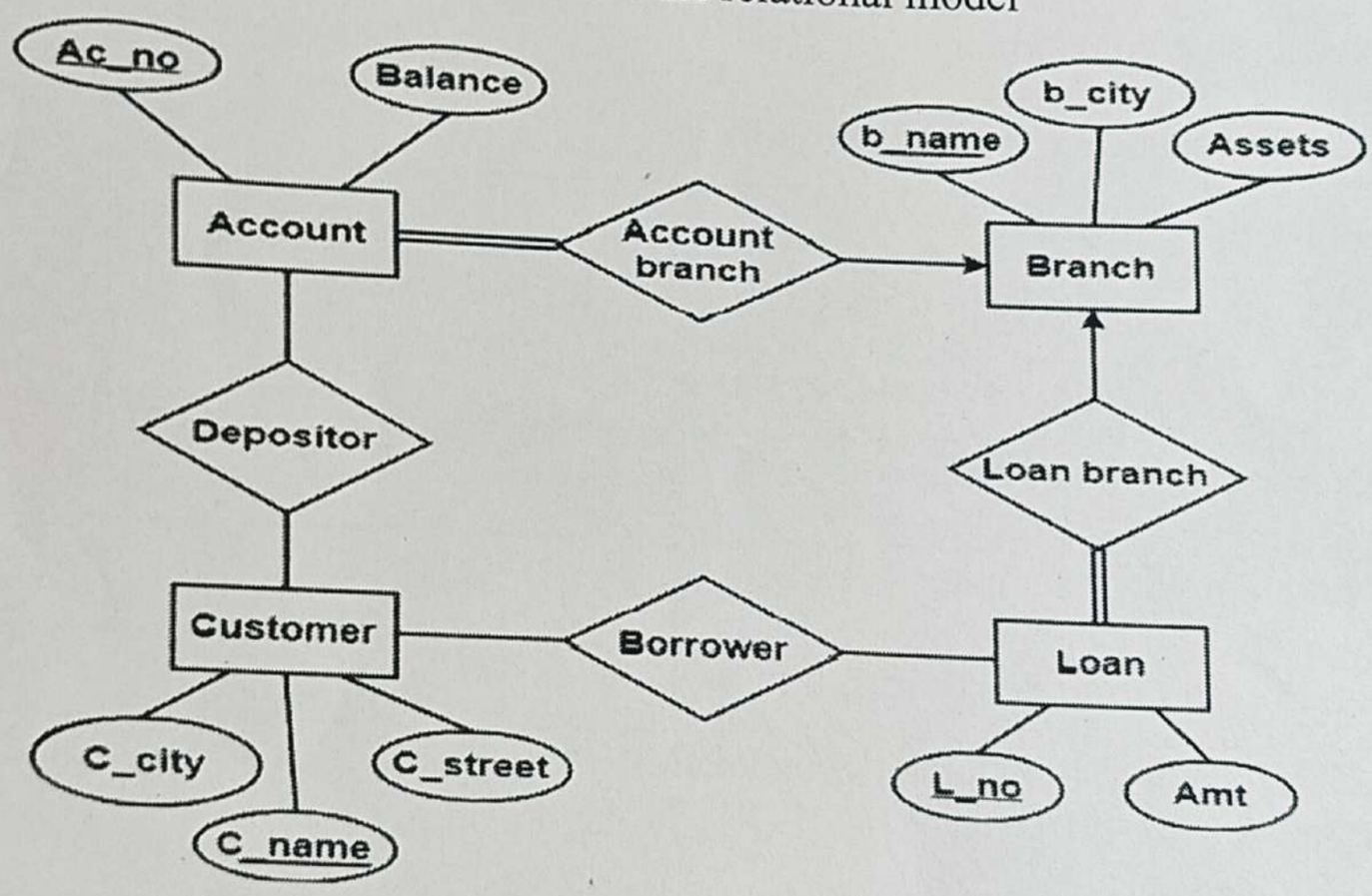
BRANCH: CSE-IOTCS

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Discuss about primary key and foreign key of a relation with example.	1	2	2
(b)	Write the role of database administrator.	1	2	2
(c)	Define the term data independence in the database.	1	2	2
(d)	Describe about composite attributes.	2	2	2
(e)	Compare and contrast strong entity with weak entity set along with example	2	2	2
(f)	List out the different types of Degrees of relationships	2	2	2
(g)	Define the term inner join and outer join.	3	2	2
(h)	Explain division operation with example.	3	2	2
(i)	Difference between drop and truncate command in SQL.	3	2	2
(j)	A relation R(A, B, C, D, E) with functional dependencies {A → BC, CD → E} is decomposed into R1(A, B, C) and R2(C, D, E). Is the decomposition lossless? Explain your answer.	4	2	3
(k)	Discuss different types of functional dependencies in a relational schema R with examples.	4	2	2
(l)	A relation R(A, B, C, D, E) with functional dependencies {A → B, BC → D, D → E} is given. Compute the closure of A and BC, and list two uses of attribute closure.	4	2	3
(m)	Differentiate between commit and rollback.	5	2	2
(n)	Make a clear distinction between serial and non-serial schedule.	5	2	2
(o)	Explain the requirements of recovery in a database environment.	5	2	2

	Part-B	CO	Marks
Q2.	**Answer any two:**		
(a)	Compare and contrast between the Relational model and Hierarchical model. Explain with examples.	1	7
(b)	i)Illustrate about the advantages of DBMS over file management system. ii) Data independence is necessary in DBMS. Justify.	1	7
(c)	Briefly explain data atomicity, isolation with data inconsistency in DBMS with examples.	1	7
Q3.	**Answer any two:**		
(a)	Illustrate Different types of keys along with examples	2	7
(b)	Describe each of the cardinality ratios and participation constraints with the help of an ER Diagram	2	7
(c)	Map the following ER model into the relational model 	2	7
Q4.	**Answer any two:**		
(a)	Consider the following relations:	3	7

P		
X	Y	Z
X1	Y1	Z1
X1	Y1	Z2
X2	Y2	Z2
X2	Y4	Z4

Q		
X	Y	T
X2	Y1	2
X1	Y2	5
X1	Y1	6
X3	Y3	1

R	
Y	V
Y1	V1
Y3	V2
Y2	V3
Y2	V2

Retrieve the tuples that will be returned by the following relational algebra query.

i) $\sigma_{P.Y=R.Y \text{ AND } R.Y=V2}(P * R)$ where * represents natural join.

ii) $\sigma_{R.Y=Q.Y}(R*Q)$ where * represents natural join

iii)Perform left outer join between Q and R.

(b)	State and explain the commands DDL, DML and DQL with suitable examples and syntax of each.	3	7	4
(c)	Consider the following relations and solve the Queries by using Relational Algebra. Employee (eid, ename, salary, dept_id) Department (dept_id, dept_name, location) i. Identify primary and foreign keys. ii. Find employees with salary greater than 40000. iii. Find employees working in 'HR' department. iv. Find departments located in 'Delhi'.	3	7	3
	Answer the followings:			
	Consider a relation R= (A, B, C, D, E, F) with two sets of functional dependencies: $F1 = \{A \rightarrow BC, B \rightarrow CDE, AE \rightarrow F\}$ $F2 = \{A \rightarrow BCF, B \rightarrow DE, E \rightarrow AB\}$ Check whether F1 and F2 are equivalent or not.	4	7	3
	Outline the necessity of normalization in a relation. Discuss about various normal forms up to BCNF with suitable examples.	4	7	3
	Answer the followings:			
	Describe the term transaction. Explain about various states of a transaction using a state diagram.	5	7	2

(b)	Define conflict serializability. Is the given schedule conflict serializable or not: If yes, justify. (i) S1: r1(x) r1(y) w2(x) w1(x) r2(y) (ii) S2: r1(x) r3(y) w1(x) w2(y) r3(x) w2(x)	5	7	3

Best of Luck

Course Outcome Assessment Scheme

COs	Questions	Total Mark
CO1: Differentiate the database concepts from conventional file storage system and describe DBMS architecture, relational, hierarchical and network database models	1(a), 1(b), 1(c), 2(a), 2(b), 2(c)	20
CO2: Be able to Analyse application data using E-R modelling and describe the logical and physical database designs.	1(d), 1(e), 1(f), 3(a), 3(b), 3(c)	20

C. V. Raman Global University
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R24

Registration No.: 2 3 0 U 0 2 0 0 2 2

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B.TECH
CS242

FOURTH SEMESTER REGULAR EXAMINATION – 2025
DATABASE ENGINEERING
CSE & Allied Branches

Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS)	CO	Marks	BL
(a)	Explain the purpose and role of a data dictionary in a database system.	1	2	1
(b)	Explain the meaning of a composite attribute in an ER model with an example.	1	2	1
(c)	Give an example of a one-to-many relationship.	1	2	2
(d)	Compare and contrast Generalization, Specialization and Aggregation.	2	2	2
(e)	Illustrate ER and EER models in the context of database.	2	2	2
(f)	Explain the concept of the degree of a relationship in an ER model and illustrate it with an example.	2	2	2
(g)	Analyse how relational algebra differs from SQL in terms of expressiveness and execution.	3	2	2
h)	Explain the difference between a JOIN operation and a Cartesian product in relational algebra.	3	2	2
i)	Explain the concept of union compatibility in the context of relational algebra operations.	3	2	
)	Identify the anomalies that occur in an un-normalized relation.	4	2	
)	Explain the process of minimizing the left-hand side of a functional dependency.	4	2	2

(l)	Identify and eliminate any redundant functional dependencies from the given set:R(ABC) $F = \{ A \rightarrow B, B \rightarrow C, A \rightarrow C \}$	4	2	3
(m)	Define the term view serializability with a suitable example.	5	2	2
(n)	Analyze the schedule **S1: R1(A), R2(B), W1(B), W2(A)** and determine whether it is conflict serializable by constructing a precedence graph.	5	2	3
(o)	List the types of failures that can occur in a database system.	5	2	2

	Part-B	CO	Marks	BL
Q2.	**Answer any two:**			
(a)	Draw and explain the three schema architecture in database design. Write the importance of data abstraction in DBMS.	1	7	2
(b)	Explain the concept of a file processing system and discuss its limitations. How does a Database Management System (DBMS) address and resolve these issues?	1	7	2
(c)	In the context of the Entity-Relationship (ER) model, explain the concept of a weak entity set. Discuss the following aspects in your answer: i) How does it differ from a strong entity set? ii) Describe the role of the identifying relationship and partial key in identifying weak entity sets. iii) Illustrate your explanation with an appropriate example (e.g., an ER diagram involving entities such as Student and Dependent, or Building and Apartment).	1	7	3
Q3.	**Answer any two:**			
(a)	Classify the different types of keys in a database and explain how each contributes to data integrity	2	7	
(b)	i) Design relational tables from a given ER diagram involving specialization. ii) Convert the given ER diagram given in figure-1 into a set of relational database tables along with mentioning the primary keys and foreign keys in each.	2	7	

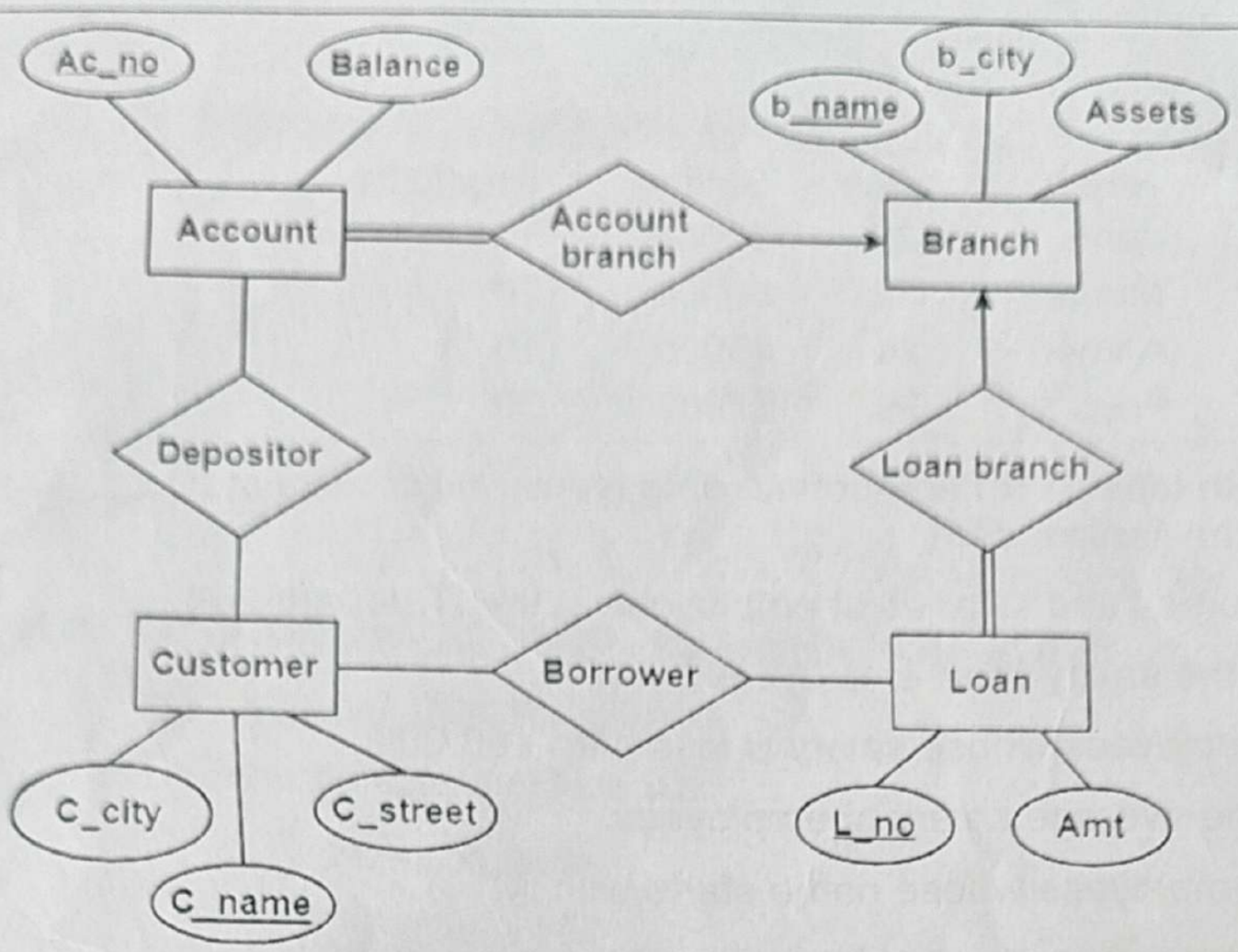


Figure-1: ER Diagram

(c)	Design an ER diagram for the following information. The company is organized into Departments. Each department has a name, number and an employee who manages the department. We keep track of the start date of the department manager. A department may have several locations. Each department controls a number of Projects. Each project has a unique name, unique number and is located at a single location. The database will store each Employee's social security number, address, salary, sex, and birthdate. Each employee works for one department but may work on several projects. The DB will keep track of the number of hours per week that an employee currently works on each project. v It is required to keep track of the direct supervisor of each employee. Each employee may have a number of Dependents. For each dependent, the DB keeps a record of name, sex, birthdate, and relationship to the employee.	2	7	4
Q4.	**Answer any two:**			
a)	Define the join operation in relational algebra. How do natural join, theta join, and equi-join differ from each other? Explain with suitable examples.	3	7	2
b)	Answer the following queries using SQL based on the given tables. **Department** DeptID | DeptName 10 | HR 20 | IT 30 | Finance	3	7	

Employee

EmpID	Name	Age	Salary	DeptID
1	John	28	50000	10
2	Maria	32	62000	20
3	Ahmed	24	45000	10
4	Priya	29	67000	30

i)Create both tables with appropriate data types and constraints and Insert the sample data.

ii)Display names and salaries of employees in the IT department

iii)Increase the salary of all employees by 10%.

iv)Delete employees whose salary is less than ₹50,000.

v) Display the average salary of employees.

vi) Display employees whose name starts with 'M'

vii) List all employees ordered by their salary from highest to lowest.

(c)	Consider the following relational database schema consisting of the four relation schemas: passenger (**pid**, pname, pgender, pcity) agency (**aid**, aname, acity) flight (**fid**, fname,fdate, time, source, destination) booking (pid, aid, fid, fdate) Answer the following questions using relational algebra queries; i)Get the flight ids of those flights travelling to New Delhi. ii) Get the flight ids and names of those flights travelling from Chennai to New Delhi. iii)Get the agency ids who booked those flights with flightid 'UA2321' iv)Find the passenger names for those passengers who have bookings on flight_id 'UA2321' v) Find the passenger names and city for those passengers who have booked flight through agency named as 'JET Agency'	3	7	3
Q5.	**Answer the followings:**			
(a)	Consider the relation R(A,B,C,D,E,F,G,H,I,J,K} with the following set of functional dependencies F given below: F= {AB→C, A→DE, B→F, F→GH, D→IJ, K→B}. Find all the candidate keys of the relation. Check whether the relation R is in BCNF? If not, normalize it up to BCNF.	4	7	